# 2A 1 STRUCTURE OF THE ATOM AND ISOTOPES

SPECIFICATION REFERENCE

2.1 2.2 2.3 2.4 2.5

## LEARNING OBJECTIVES

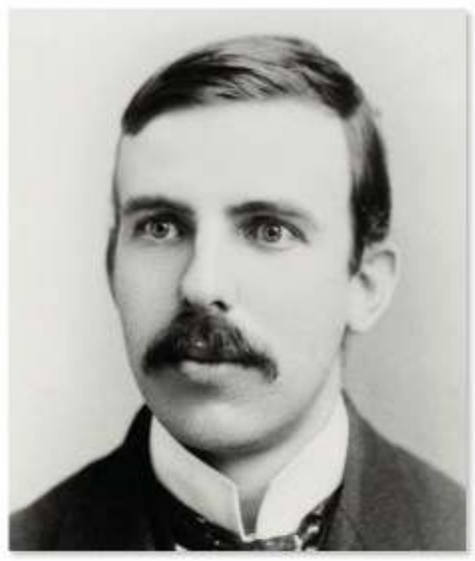
- Know the structure of an atom in terms of electrons, protons and neutrons.
- Know the relative mass and charge of protons, neutrons and electrons.
- Know the meaning of the terms atomic (proton) number and mass number.
- Be able to use the atomic number and the mass number to determine the number of each type of subatomic particle in an atom or ion.
- Understand the term isotope.

## WHO DISCOVERED ELECTRONS, PROTONS AND NEUTRONS?

(a)



(b)



(c)



▲ **fig A** (a) J.J. Thomson, (b) Ernest Rutherford and (c) James Chadwick.

Our current understanding of the structure of atoms is influenced by the theories put forward by scientists such as J.J. Thomson, Ernest Rutherford and James Chadwick.

J.J. Thomson discovered the electron in 1897. Ernest Rutherford discovered the proton in 1917. James Chadwick discovered the neutron in 1932.

## STRUCTURE OF AN ATOM

Although scientists have discovered many other subatomic particles, chemistry is only concerned with electrons, protons and neutrons.

We can summarise the structure of the atom in terms of these three subatomic particles, as shown in **table A**.

| PARTICLE | SYMBOL | RELATIVE MASS | RELATIVE CHARGE | POSITION IN THE ATOM |
|---|---|---|---|---|
| proton | p | 1 | +1 | nucleus |
| neutron | n | 1 | 0 | nucleus |
| electron | $e^-$ | $\frac{1}{1840}$ | −1 | energy levels surrounding the nucleus |

**table A** The structure of the atom in terms of protons, neutrons and electrons.

Rutherford discovered the proton and he is also credited for first suggesting that the atom has a very small core containing the bulk of the mass of the atom. This core is called the nucleus and contains all of the protons and neutrons in that atom.

Electrons are also present in atoms. The electrons exist in energy levels surrounding the nuclei. The energy levels are called quantum shells. We will develop this concept throughout **Topic 2A**.

### LEARNING TIP

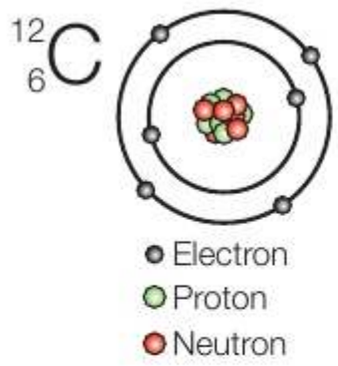
Never state that the mass of an electron is zero. This is not true, an electron does have a mass.
You do not need to know the exact masses of subatomic particles in grams, or the exact charges in coulombs. You need to know the relative values.

### DID YOU KNOW?

All atoms of any element have the same atomic number, which is different from the atomic number of any other element.
The number of electrons in a neutral atom of an element is equal to the number of protons in the nucleus of that atom.
The electrons surround the nucleus in well-defined energy levels called quantum shells.

## ATOMIC NUMBER, MASS NUMBER AND ISOTOPES

**Fig B** shows the **atomic number**, **mass number** and **isotopes** of carbon.




**Carbon atoms**

Mass number → $^{12}_{6}C$, $^{13}_{6}C$ ← Atomic number

Atomic number = number of protons (= number of electrons)

Mass number = number of protons + number of neutrons

▲ **fig B** Atomic number, mass number and isotopes of carbon.

Two isotopes of carbon are $^{12}C$ and $^{13}C$. They have the same atomic number, 6, and the same electronic configuration, 2.4, (see **Topic 2A.3**), so samples of carbon containing either isotope will have the same chemical properties. However, the mass of $^{13}C$ is larger than $^{12}C$. This is because $^{13}C$ has one more neutron.

## ISOTOPES

In 1919, Francis Aston invented an instrument called a mass spectrometer. He used it to discover that not all the atoms of an element have the same mass. It was eventually realised that the different masses were caused by the atoms having different numbers of neutrons.

These different atoms are called isotopes.

We can illustrate this point by considering two isotopes of chlorine: chlorine-35 and chlorine-37.

**Table B** shows the number of protons, neutrons and electrons in an atom of each isotope.

| ISOTOPE | SYMBOL FOR ISOTOPE | NUMBER OF PROTONS | NUMBER OF NEUTRONS | NUMBER OF ELECTRONS |
|---|---|---|---|---|
| chlorine-35 | $^{35}_{17}Cl$ | 17 | 18 | 17 |
| chlorine-37 | $^{37}_{17}Cl$ | 17 | 20 | 17 |

**table B** Isotopes of chlorine.

**DID YOU KNOW?**

Isotopes of the same element have identical chemical properties because they have identical electronic configurations.

## CHECKPOINT

**1.** Use the information in the table below to answer the following questions about particles **A** to **F**.

(a) Which two particles are isotopes of the same element?

(b) Which two particles are positive ions?

(c) Which two particles are negative ions?

(d) Which two particles have the same mass number?

| PARTICLE | NUMBER OF PROTONS | NUMBER OF NEUTRONS | NUMBER OF ELECTRONS |
|---|---|---|---|
| **A** | 12 | 13 | 12 |
| **B** | 17 | 18 | 18 |
| **C** | 11 | 14 | 10 |
| **D** | 12 | 12 | 12 |
| **E** | 35 | 44 | 36 |
| **F** | 19 | 21 | 18 |

**2.** Complete the table below to show the numbers of protons, neutrons and electrons in these atoms, molecules and ions.

| SYMBOL FOR ATOM OR ION | NUMBER OF PROTONS | NUMBER OF NEUTRONS | NUMBER OF ELECTRONS |
|---|---|---|---|
| $^{3}_{1}H$ | | | |
| $^{18}_{8}O^{2-}$ | | | |
| $^{24}_{12}Mg^{2+}$ | | | |
| $^{14}_{7}N^{1}_{1}H_3$ | | | |
| $^{14}_{7}N^{2}_{1}H_4^{+}$ | | | |

## SUBJECT VOCABULARY

**atomic number (*Z*)** the number of protons in the nucleus of an atom

**mass number** the sum of the number of protons and the number of neutrons in the nucleus of an atom

**isotopes** atoms of the same element that have the same atomic number but different mass number

2A

# 2 MASS SPECTROMETRY AND RELATIVE MASSES OF ATOMS, ISOTOPES AND MOLECULES

SPECIFICATION REFERENCE: 2.6 2.7

## LEARNING OBJECTIVES

- Understand the basic principles of a mass spectrometer and be able to analyse and interpret spectra to: deduce the isotopic composition of an element; calculate relative atomic mass from the relative abundance of isotopes and vice versa; determine the relative molecular mass of a molecule, and identify molecules in a sample.
- Understand that ions in a mass spectrometer may have a 2+ charge.
- Be able to predict the mass spectra, including relative peak heights, for diatomic molecules including chlorine, given the isotopic abundances.

## RELATIVE ATOMIC MASS AND RELATIVE ISOTOPIC MASS

### RELATIVE ATOMIC MASS

Chemists working in the 19th century established the chemical formulae of some common compounds. They also calculated the masses of atoms from which these compounds are often made. In fact, they hadn't measured the actual masses of the atoms but how the masses of atoms compared with one another, in other words **relative atomic masses**.

The sulfur atom was twice as heavy as the oxygen atom, and the oxygen atom was sixteen times heavier than the hydrogen atom. The hydrogen atom was the lightest atom, so its mass was given a value of 1. The masses of the atoms of the other elements were stated relative to this value. So the mass of an atom of oxygen on this scale was 16, and sulfur was 32.

The discovery of isotopes complicated the situation. In 1961, scientists decided to adopt an isotope of carbon, carbon-12, as the standard.

### RELATIVE ISOTOPIC MASS

**Relative isotopic masses** and their percentage abundance in a sample of an element are used to calculate relative atomic masses.

#### WORKED EXAMPLE

Lithium, in its naturally occurring compounds, has two isotopes of relative isotopic masses 6.015 and 7.016. The percentage abundance of each isotope is 7.59 and 92.41 respectively.

This is how you can calculate the relative atomic mass of lithium.

There are two stages.

Stage 1: $(6.015 \times 7.59) + (7.016 \times 92.41) = 694.00241$

Stage 2: $\frac{694.00241}{100} = 6.9400241$

The relative atomic mass of lithium, to three significant figures, is 6.94.

#### EXAM HINT

The relative atomic mass will always be a number somewhere between the relative isotopic masses. If your calculator gives you a different number, you have made a mistake and should go back and check your workings.

#### LEARNING TIP

Remember that:

- the Z number is not the same as the relative isotopic mass; the mass number is always a whole number because it is the sum of two whole numbers (the number of protons and the number of neutrons in the nucleus of an atom)
- relative isotopic mass is relative to the mass of a carbon-12 atom; it is not likely to be a whole number
- relative isotopic masses should be used to calculate relative atomic masses, but sometimes mass numbers are used to make the arithmetic easier.

## USING DATA OBTAINED FROM A MASS SPECTROMETER

### WHAT IS A MASS SPECTROMETER?

A mass spectrometer measures the masses of atoms and molecules.

It produces positive ions that are deflected by a magnetic field according to their mass-to-charge ratio ($m/z$). It also calculates the relative abundance of each positive ion and displays this as a percentage.

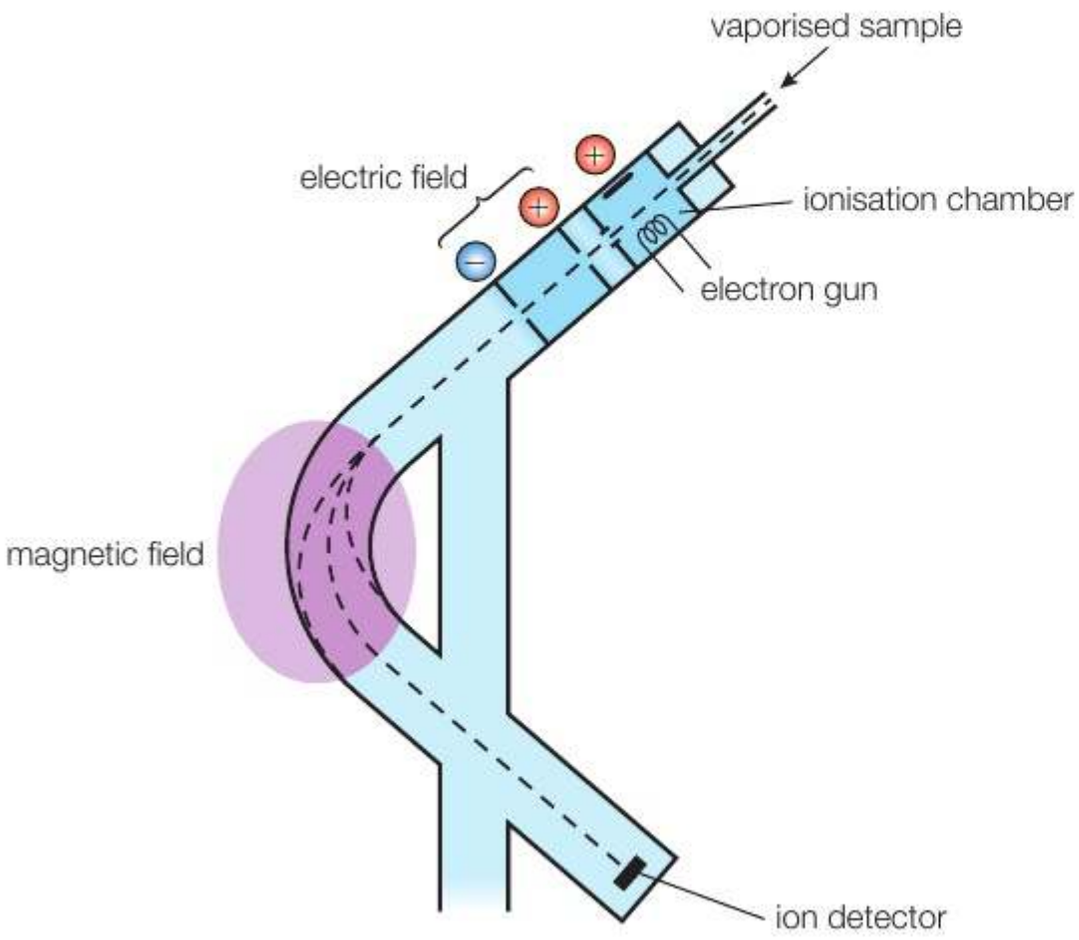
The positive ions could be positively charged atoms, positively charged molecules or positively charged fragments of molecules.

#### DID YOU KNOW?

**The relative atomic mass of an element is the weighted mean mass of an atom of the element compared to $\frac{1}{12}$ the mass of an atom of carbon-12, which has a mass of 12.**
**The relative isotopic mass is the mass of an atom of an isotope of the element compared to $\frac{1}{12}$ the mass of an atom of carbon-12, which has a mass of 12.**

Dealing with fragments of molecules is particularly useful when you are analysing organic compounds. We will discuss this in more detail in **Topic 10**.

## HOW DOES A MASS SPECTROMETER WORK?




▲ **fig A** A mass spectrometer.

1. The sample being analysed must be in the gaseous state so that its particles can move through the machine. The sample is injected into the mass spectrometer, where it is vaporised.
2. The vapour is bombarded with high energy electrons. These electrons collide with the atoms or molecules of the sample. One or more electrons are removed from the atoms or molecules to form positive ions.
3. An electric field causes the positive ions to accelerate.
4. The positive ions then enter, and are deflected by, a uniform magnetic field. The amount of deflection depends on the mass-to-charge ($m/z$) ratio of the ions. Ions with a large mass-to-charge ratio are deflected less than ions with a small mass-to-charge ratio. If all the ions have the same charge, usually 1+, the extent of the deflection is inversely proportional to their mass.
5. The deflected ions pass through a narrow slit and are collected on a metallic plate connected to an amplifier. For a given strength of magnetic field, only ions with a certain $m/z$ ratio pass through the slit and are detected. For example, an ion with a mass of 28 and a charge of 1+ will be detected at the same time as an ion with a mass of 56 and a charge of 2+. Both ions have a $m/z$ ratio of 28.

   The strength of the magnetic field is then changed to detect positive ions with other $m/z$ ratios.

## DETERMINING RELATIVE ISOTOPIC AND ATOMIC MASSES

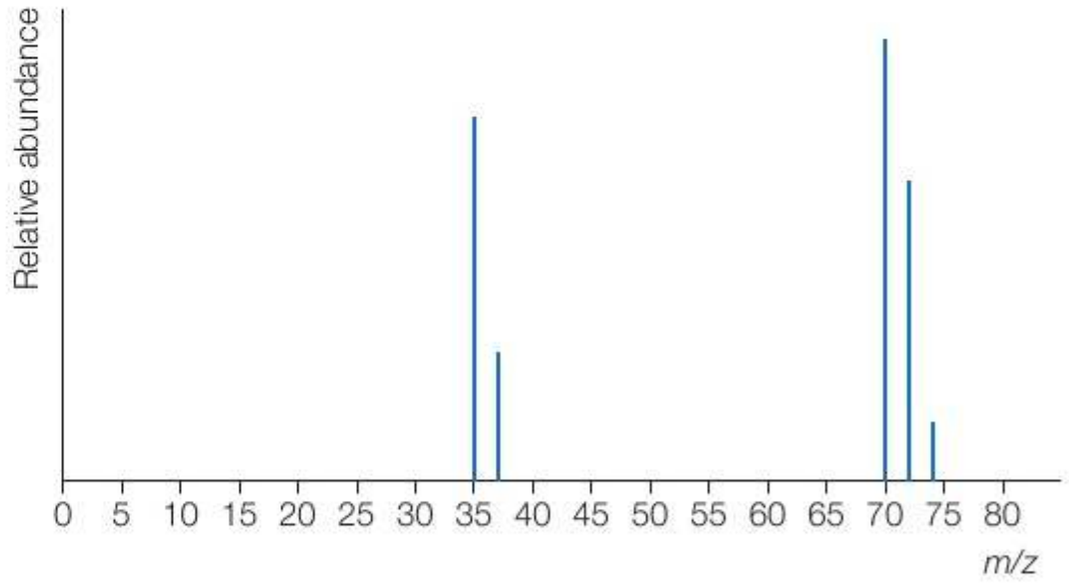
You can determine the *exact* values of the relative masses of isotopes from a mass spectrum of an element, together with the percentage abundance of each isotope.

You can use this information to calculate the relative atomic mass of the element.

## DETERMINING RELATIVE MOLECULAR MASS ($M_r$) OF DIATOMIC MOLECULES

Some elements and compounds contain two or more atoms covalently bonded together. If these substances are analysed by mass spectrometry, you can obtain the relative molecular mass of the element or compound by observing the peaks with the largest $m/z$ ratios (assuming a value of $z = 1$).

**Fig B** shows the mass spectrum of chlorine, which exists as diatomic molecules: $Cl_2$.




▲ **fig B** Mass spectrum of chlorine.

To make calculations easier, in this example the relative molecular and isotopic masses are quoted as whole numbers.

There are two peaks corresponding to isotopic masses of 35 and 37. The approximate relative peak heights for each isotope are 3 : 1 for chlorine-35 : chlorine-37.

This gives an approximate relative atomic mass of this sample of chlorine of $((3 \times 35) + (1 \times 37)) \div 4 = 35.5$.

You will notice in the diagram that there are three peaks, which correspond to molecular masses of 70, 72 and 74. How can we explain the relative heights of these peaks?

**Table A** shows the atomic composition of each molecule of chlorine.

| MASS OF MOLECULE | FORMULA OF MOLECULE |
|---|---|
| 70 | $^{35}Cl^{35}Cl$ |
| 72 | $^{35}Cl^{37}Cl$ |
| 74 | $^{37}Cl^{37}Cl$ |

**table A** Atomic composition of each molecule of chlorine.

The ratio of $^{35}Cl$ to $^{37}Cl$ in this sample is 3 : 1. The molecular composition is shown in **table B**.

| FORMULAE OF MOLECULE | RATIO OF MOLECULE |
|---|---|
| $^{35}Cl^{35}Cl$ | 9 |
| $^{35}Cl^{37}Cl$ | 6 |
| $^{37}Cl^{37}Cl$ | 1 |

**table B** Molecular composition of sample.

**LEARNING TIP**

The peak heights will not be exactly 9 : 6 : 1 because the relative isotopic masses are not whole numbers.

Let's explain this.

- There is a 3 in 4 chance of selecting a $^{35}Cl$ atom from a sample of chlorine atoms. This means that the total chance of two $^{35}Cl$ atoms combining together is $\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$.
- There is a 1 in 4 chance of selecting a $^{37}Cl$ atom. This means that the chance of $^{37}Cl$ combining with a $^{35}Cl$ is $\frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$. The chance of a $^{35}Cl$ atom combining with a $^{37}Cl$ atom is also $\frac{3}{16}$. This means that the total chance of $^{35}Cl$ and $^{37}Cl$ combining together in any order is $2 \times \frac{3}{16}$ or $\frac{6}{16}$.
- The chance of two $^{37}Cl$ atoms combining together is $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$

  $\frac{9}{16} : \frac{6}{16} : \frac{1}{16}$ in whole number ratios is 9 : 6 : 1.

This corresponds approximately to the peak heights of 70, 72 and 74.

**LEARNING TIP**

Make sure you always use the relative atomic masses you are given in any question. For example, the value for magnesium given in the Periodic Table on page 306 at the end of this book is 24.3. However, you may occasionally be asked to use a more accurate value, such as 24.305.

You can then determine the relative molecular mass ($M_r$) of chlorine by calculating the weighted mean of the various molecules present:

$$M_r(Cl_2) = \frac{(9 \times 70) + (6 \times 72) + (1 \times 74)}{16} = 71$$

You will usually be asked to calculate relative molecular masses by simply adding together the relative atomic masses of the elements in the molecule.

If we use this method, the relative molecular mass of chlorine is $2 \times 35.5 = 71$.

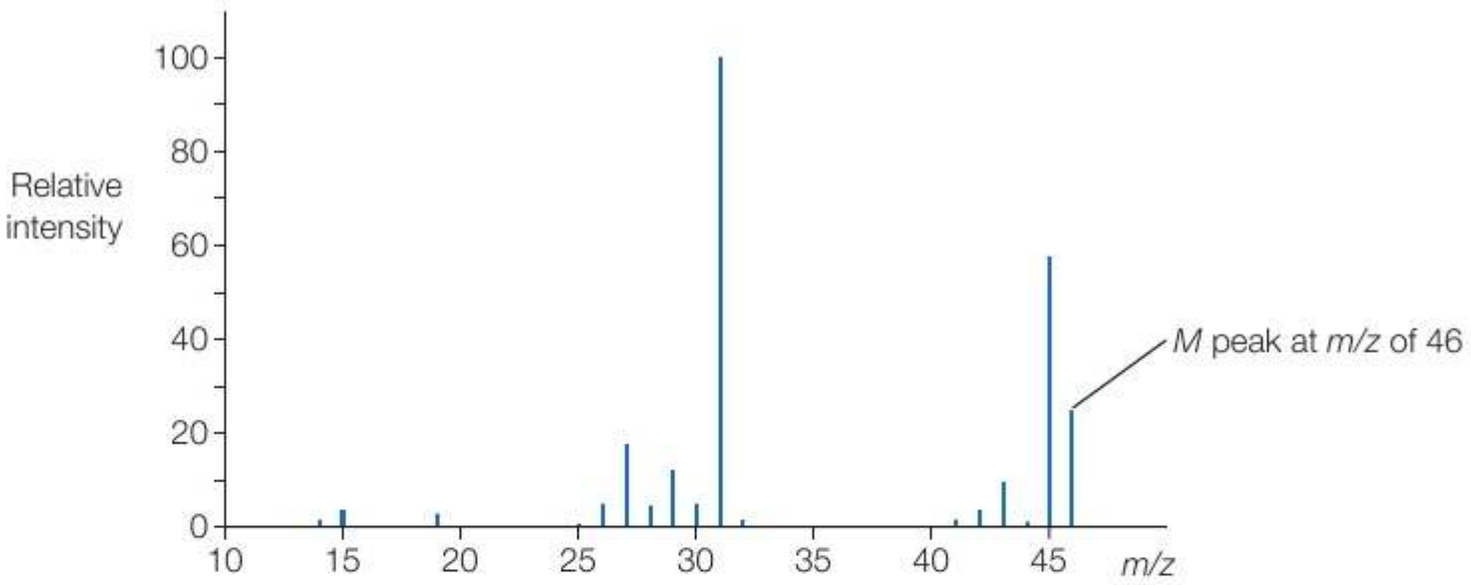
## DETERMINING RELATIVE MOLECULAR MASS OF A POLYATOMIC MOLECULE

It is quite difficult to use the data supplied by a mass spectrum of a compound to determine its exact relative molecular mass. You need to know the exact relative isotopic masses of all the atoms present, and the relative composition of all the different molecules.

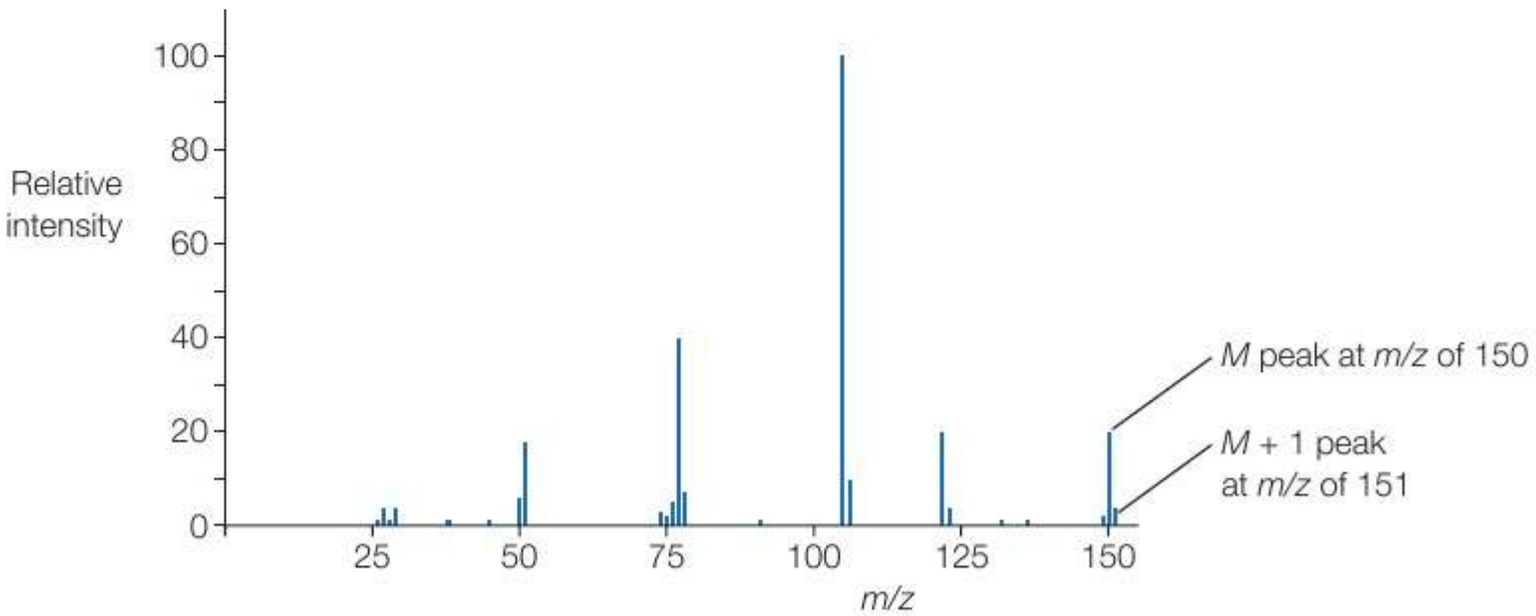
You are more likely to be asked to work out the relative molecular mass of a compound by considering what is called the **molecular ion peak**.

You should be careful when analysing organic compounds. This is because there is always a small percentage of the carbon-13 isotope present in the compound, which can lead to what is referred to as an $M+1$ peak. This peak can often be seen in molecules with large masses, where the percentage of carbon-13 becomes significant. The peak is often missing, or insignificant, in molecules of small mass.

The following examples illustrate this point.




▲ **fig C** Mass spectrum of ethanol ($M_r$ = 46).

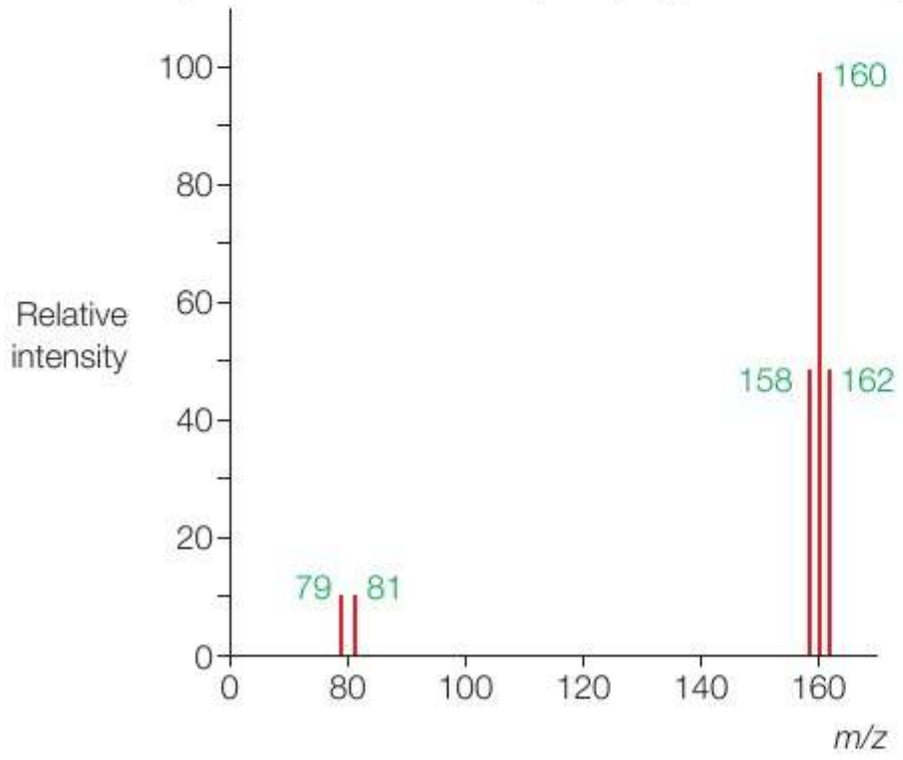



▲ **fig D** Mass spectrum of ethyl benzoate ($M_r$ = 150).

## CHECKPOINT

1. Describe the difference between relative atomic mass and relative isotopic mass.
2. Explain why the relative atomic masses of many elements are not exact whole numbers.
3. Calculate the relative atomic mass of a sample of magnesium that has the following isotopic composition:
   - magnesium-24: 78.6%
   - magnesium-25: 10.1%
   - magnesium-26: 11.3%

   Give your answer to three significant figures.
4. A sample of copper contains two isotopes of relative isotopic mass 63.0 and 65.0. If the relative atomic mass of copper is 63.5, calculate the relative abundance of each isotope.
5. The mass spectrum of bromine vapour, $Br_2$, is shown in **fig E** below.




▲ **fig E**

(a) What are the relative isotopic masses of the two isotopes present in bromine?

(b) Identify the particles responsible for the peaks at *m/z* 158, 160 and 162.

(c) Deduce the relative abundance of the two isotopes and explain the relative heights of the three peaks at *m/z* 158, 160 and 162.

SKILLS REASONING

### EXAM HINT

In exam questions, always give your answer to the number of significant figures asked for, otherwise you will probably lose a mark.

## SUBJECT VOCABULARY

**relative atomic mass ($A_r$)** (of an element) the weighted mean (average) mass of an atom of the element compared to $\frac{1}{12}$ of the mass of an atom of carbon-12

**relative isotopic mass** the mass of an individual atom of a particular isotope relative to $\frac{1}{12}$ of the mass of an atom of carbon-12

**molecular ion peak** the peak with the highest *m/z* ratio in the mass spectrum, the *M* peak

# 2A 3 ATOMIC ORBITALS AND ELECTRONIC CONFIGURATIONS

SPECIFICATION REFERENCE

2.9 2.12 2.13 2.14 2.15

## LEARNING OBJECTIVES

- Know that an orbital is a region within an atom that can hold up to two electrons with opposite spins.
- Be able to describe the shapes of s and p orbitals.
- Know that orbitals in sub-shells: each take a single electron before pairing up; pair up with two electrons of opposite spin.
- Be able to predict the electronic configuration of the atoms of the elements from hydrogen to krypton inclusive and their ions, using s, p, d notation and electron-in-boxes notation.
- Understand that electronic configuration determines the chemical properties of an element.
- Know the number of electrons that can fill the first four quantum shells.
- State the number of electrons that occupy s, p and d sub-shells.

## QUANTUM SHELLS

Max Planck first presented the quantum theory in 1900. We can use this theory to describe the arrangement of electrons around the nuclei of atoms.

According to this theory, electrons can only exist in certain well-defined energy levels called **quantum shells**.

All electrons in a quantum shell have similar, but not identical, energies.

### ELECTRONS IN THE FIRST FOUR QUANTUM SHELLS

Electrons in the first quantum shell have the lowest energy for that element. The first quantum shell is found in the region closest to the nucleus. The second quantum shell is in a region outside the first; the third is outside the second, and so on.

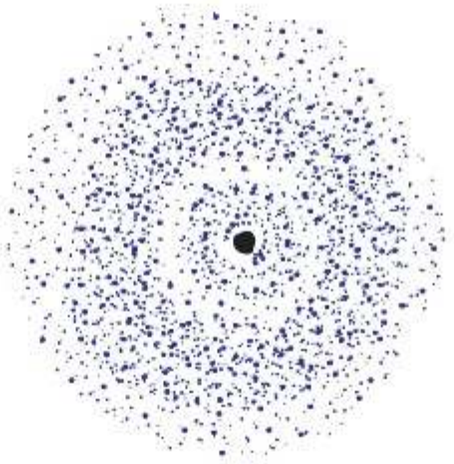
Each quantum shell, apart from the first, is further divided into sub-shells of slightly different energy levels.

- There is only one sub-shell in the first quantum shell. This is labelled the 1s sub-shell.
- The second quantum shell has two sub-shells, labelled 2s and 2p. Electrons in the 2p sub-shell have a slightly higher energy level than those in the 2s sub-shell.
- The third quantum shell is divided into three sub-shells, labelled 3s, 3p and 3d. Electrons in the 3p sub-shell have slightly higher energy than those in 3s, and those in the 3d sub-shell have slightly higher energy than those in 3p.
- The fourth quantum shell is divided into four sub-shells, labelled 4s, 4p, 4d and 4f. Again, the electron energies increase in the order 4s < 4p < 4d < 4f.

### s ORBITALS

Each sub-shell contains **orbitals**. If you could see a single electron in a hydrogen atom and map its position at regular intervals, then you could construct a three-dimensional map of places where the electron is likely to be found. Most of the time, the electron is located within a fairly easily-defined region of space close to the nucleus.

**Fig A** shows a cross-section through a spherical s orbital.

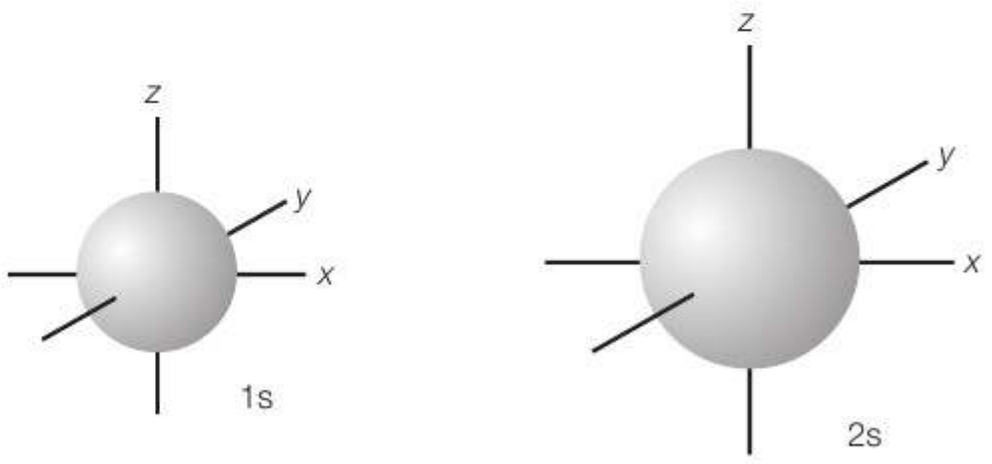


▲ **fig A** Cross-section through an s orbital.

Because the electron of the hydrogen atom is in an s sub-shell, the orbital is described as a *1s orbital*.

Electrons in the 2s sub-shell also exist in an s orbital, called the 2s orbital. The 2s orbital has the same shape as the 1s orbital but is larger. **Fig B** shows the shape of typical 1s and 2s orbitals. Note that *x*, *y* and *z* are 3D Cartesian axes (i.e. axes at mutual right angles).

The sizes and shapes of the orbitals mean that there is a 90% probability of finding the electron within their boundaries.

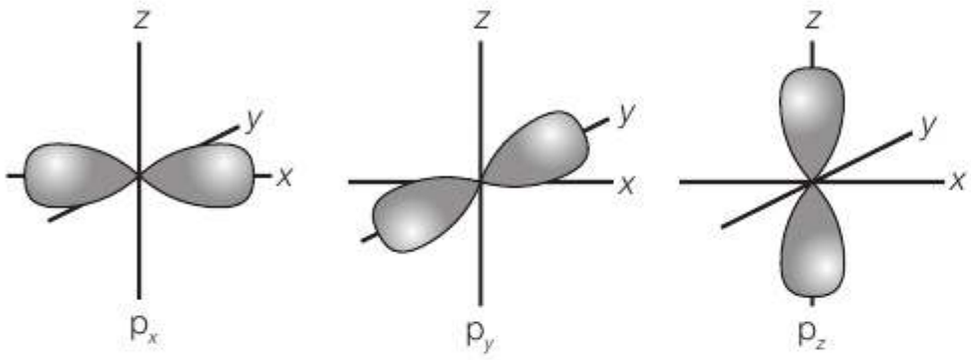



▲ **fig B** The shape of typical 1s and 2s orbitals.

## p ORBITALS

The 2p sub-shell contains three separate p orbitals. These orbitals are more complex and harder to visualise. They have an elongated dumbbell shape and a variable charge density. The only difference between the three orbitals is their orientation in space, as shown in **fig C**.

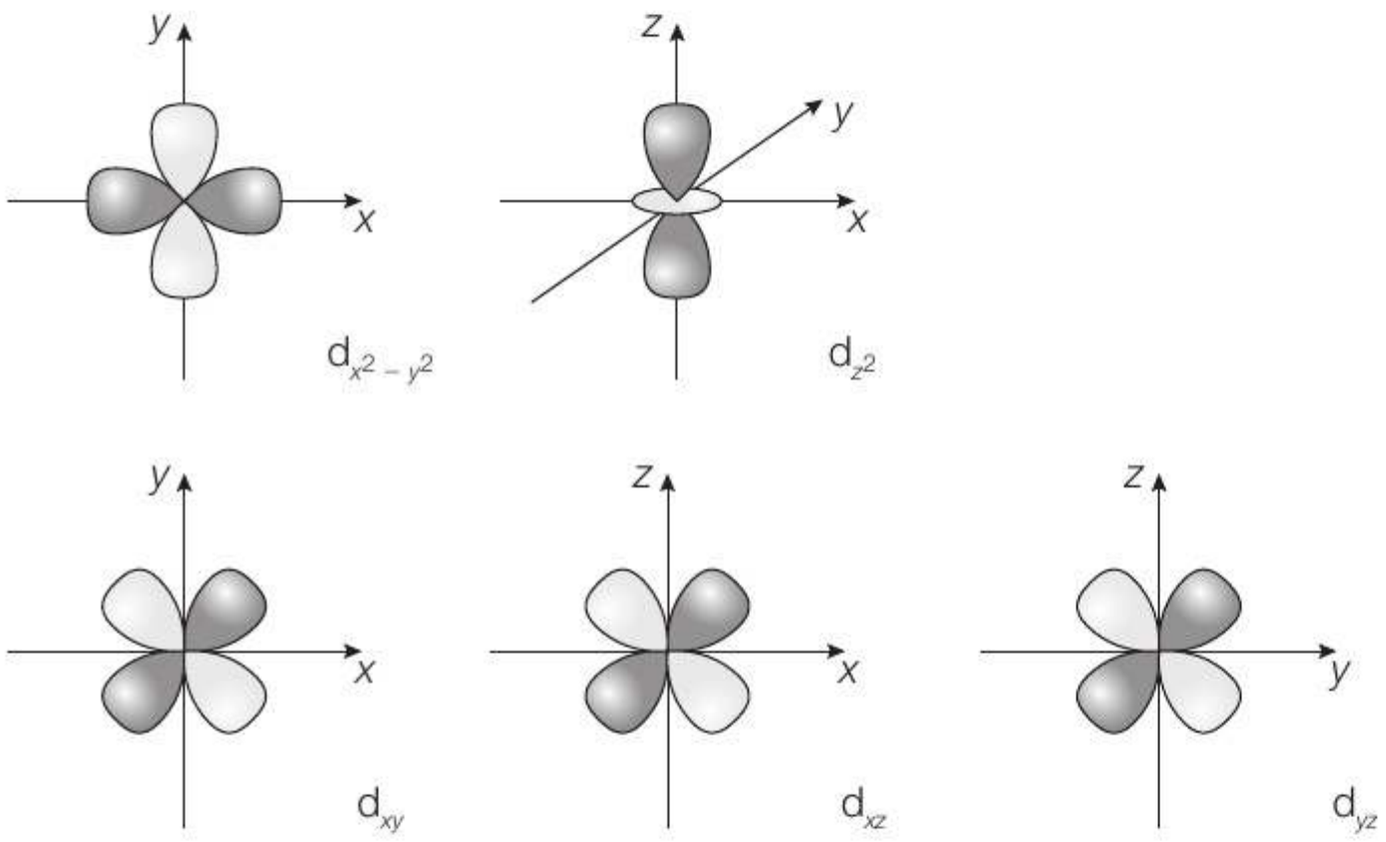
Again, the size and shape of the orbitals means that there is a 90% probability of finding the electron within their boundaries. The three orbitals are arranged at mutual right angles.




▲ **fig C** Orientation of the three p orbitals.

## d ORBITALS

The d orbitals are very complicated. You do not need to know their shapes for IAS or IAL studies. There are five d orbitals in the d sub-shell.




▲ **fig D** The five d orbitals.

There can be up to two electrons in each orbital. Using this information, the number of electrons that can exist in each sub-shell and each quantum shell is shown in **table A**.

| | NUMBER OF ELECTRONS |
|---|---|
| s sub-shell | 2 (1 × 2) |
| p sub-shell | 6 (3 × 2) |
| d sub-shell | 10 (5 × 2) |
| f sub-shell | 14 (7 × 2) |

| | NUMBER OF ELECTRONS |
|---|---|
| first quantum shell | 2 |
| second quantum shell | 8 |
| third quantum shell | 18 |
| fourth quantum shell | 32 |

**table A** The number of electrons in each sub-shell and each quantum shell.

## ELECTRONIC CONFIGURATIONS

The **electronic configuration** of an atom of an element is the distribution of electrons among atomic orbitals.

**Table B** shows the electronic configurations for the first 36 elements in the Periodic Table.

| ATOMIC NUMBER | SYMBOL | 1s | 2s | 2p | 3s | 3p | 3d | 4s | 4p |
|---|---|---|---|---|---|---|---|---|---|
| 1 | H | 1 | | | | | | | |
| 2 | He | 2 | | | | | | | |
| 3 | Li | 2 | 1 | | | | | | |
| 4 | Be | 2 | 2 | | | | | | |
| 5 | B | 2 | 2 | 1 | | | | | |
| 6 | C | 2 | 2 | 2 | | | | | |
| 7 | N | 2 | 2 | 3 | | | | | |
| 8 | O | 2 | 2 | 4 | | | | | |
| 9 | F | 2 | 2 | 5 | | | | | |
| 10 | Ne | 2 | 2 | 6 | | | | | |
| 11 | Na | 2 | 2 | 6 | 1 | | | | |
| 12 | Mg | 2 | 2 | 6 | 2 | | | | |
| 13 | Al | 2 | 2 | 6 | 2 | 1 | | | |
| 14 | Si | 2 | 2 | 6 | 2 | 2 | | | |
| 15 | P | 2 | 2 | 6 | 2 | 3 | | | |
| 16 | S | 2 | 2 | 6 | 2 | 4 | | | |
| 17 | Cl | 2 | 2 | 6 | 2 | 5 | | | |
| 18 | Ar | 2 | 2 | 6 | 2 | 6 | | | |
| 19 | K | 2 | 2 | 6 | 2 | 6 | | 1 | |
| 20 | Ca | 2 | 2 | 6 | 2 | 6 | | 2 | |
| 21 | Sc | 2 | 2 | 6 | 2 | 6 | 1 | 2 | |
| 22 | Ti | 2 | 2 | 6 | 2 | 6 | 2 | 2 | |
| 23 | V | 2 | 2 | 6 | 2 | 6 | 3 | 2 | |
| 24 | Cr | 2 | 2 | 6 | 2 | 6 | 5 | 1 | |
| 25 | Mn | 2 | 2 | 6 | 2 | 6 | 5 | 2 | |
| 26 | Fe | 2 | 2 | 6 | 2 | 6 | 6 | 2 | |
| 27 | Co | 2 | 2 | 6 | 2 | 6 | 7 | 2 | |
| 28 | Ni | 2 | 2 | 6 | 2 | 6 | 8 | 2 | |
| 29 | Cu | 2 | 2 | 6 | 2 | 6 | 10 | 1 | |
| 30 | Zn | 2 | 2 | 6 | 2 | 6 | 10 | 2 | |
| 31 | Ga | 2 | 2 | 6 | 2 | 6 | 10 | 2 | 1 |
| 32 | Ge | 2 | 2 | 6 | 2 | 6 | 10 | 2 | 2 |
| 33 | As | 2 | 2 | 6 | 2 | 6 | 10 | 2 | 3 |
| 34 | Se | 2 | 2 | 6 | 2 | 6 | 10 | 2 | 4 |
| 35 | Br | 2 | 2 | 6 | 2 | 6 | 10 | 2 | 5 |
| 36 | Kr | 2 | 2 | 6 | 2 | 6 | 10 | 2 | 6 |

**table B**

As you can see, in general the lowest energy orbitals are occupied and filled first.

For example, the electronic configuration of helium (He) is $1s^2$ not $1s^12s^1$. This is because the 1s orbital can accommodate two electrons, and electrons in the 1s orbital have a lower energy than those in the 2s orbital.

## PREDICTING ELECTRONIC CONFIGURATIONS

However, as you can see from **table B**, there are some exceptions to this general rule.

For example, the electronic configuration of potassium (K) is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ and not, as you might expect, $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$.

This is because the exact energies of the electrons in the orbitals are determined by the number of protons in the nucleus of the atom and the repulsion between all of the electrons present in the atom. For both the potassium atom and the calcium atom, the energy of the 4s orbital is lower than that of the 3d orbitals.

This means that the electronic configuration of calcium (Ca) is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ and *not* $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2$.

**DID YOU KNOW?**

**Interestingly, for all the elements after calcium in the Periodic Table, the energy of the 3d orbitals is *less* than that of the 4s orbitals.**

**This means that you would expect the electronic configuration of scandium (Sc) to be $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$.**

**However, it is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$.**

**As Professor Eric Scerri of the University of California indicated, the 3d orbitals accept electrons before the 4s orbitals: so why are two electrons pushed into the higher-energy 4s orbital? The reason is that 3d orbitals are more compact than the 4s orbitals and, as a result, any electrons entering the 3d orbital will experience greater mutual repulsion than they would in the 4s orbital. When scandium ionises, it is the 4s electrons, i.e. those with the greater energy, that are lost first. This is what happens to the elements from scandium to zinc. We will discuss this in more detail in Topic 17 (Book 2: IAL).**

**However, other atoms do not always act in this way. The nuclear charge increases as we move from scandium to zinc. This complicates the interactions between the electrons and the nucleus, and between the electrons themselves. This is what ultimately produces an electronic configuration. Unfortunately, there is not one simple rule we can use in this complicated situation.**

**For example, you might think, incorrectly, that the most stable electronic configuration for chromium (Cr) is $3d^4 4s^2$ and for copper (Cu) is $3d^9 4s^2$. This is not the case, as we have shown in the table of electronic configurations (table B).**

**LEARNING TIP**

You may have heard of a principle called the 'Aufbau principle'. Chemists often quote this principle as a method for working out the electronic configuration of the atoms. According to the Aufbau principle, as protons are added to the nucleus, the electrons are successively added to orbitals of increasing energy. This begins with the lowest energy orbitals, and continues until all electrons are accommodated. It can be a useful general rule for working out the configurations of the first 36 elements, with the exceptions of chromium and copper. However, as you study more elements you will discover more exceptions. You can use the Aufbau principle to work out electronic configurations, but it does not offer an explanation for them.

## SPIN-SPIN PAIRING AND ELECTRONIC CONFIGURATIONS

### HUND'S RULE AND THE PAULI EXCLUSION PRINCIPLE

**Hund's rule** states that electrons will occupy the orbitals singly before pairing takes place.

The **Pauli Exclusion Principle** states that two electrons cannot occupy the same orbital unless they have opposite spins.

Hund's rule and the Pauli Exclusion Principle are responsible for the way in which electrons fill the p and the d sub-shells.

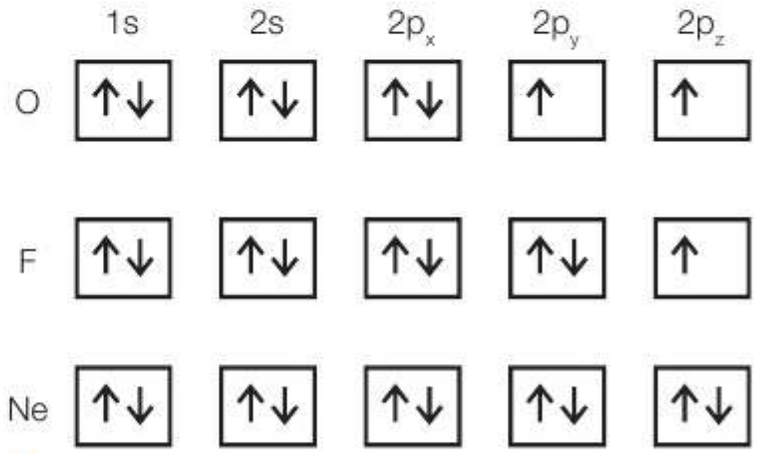
If we apply Hund's rule to boron (B), carbon (C) and nitrogen (N), we obtain the following electronic configurations:

B $1s^2 2s^2 2p_x^1$

C $1s^2 2s^2 2p_x^1 2p_y^1$

N $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$

If we apply the Pauli Exclusion Principle to oxygen (O), fluorine (F) and neon (Ne), the electronic configurations are displayed as boxes.



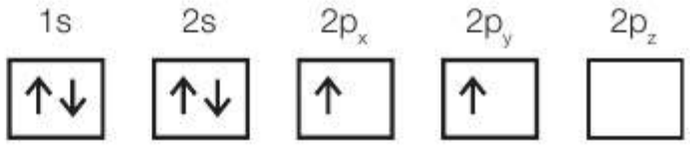

▲ **fig E** Electronic configurations for oxygen, fluorine and neon displayed in boxes.

## CHECKPOINT

**1.** The way that electrons of an atom are located in atomic orbitals is subject to Hund's rule and the Pauli Exclusion Principle.

(a) State what you understand by Hund's rule and the Pauli Exclusion Principle.

(b)




▲ **fig F** Box notation diagram for Question 1 (b) and (c).

(i) State the significance of the arrows and of the letters *x*, *y* and *z* in **fig F**.

(ii) Suggest why the electron in the $2p_y$ orbital is not located in the $2p_x$ orbital.

(iii) Describe the shapes of s and p orbitals.

(c) Using the box notation shown above, give the electronic configuration of:

(i) sulfur

(ii) phosphorus

(iii) chromium.

## SUBJECT VOCABULARY

**quantum shell** the energy level of an electron

**orbital** a region within an atom that can hold up to two electrons with opposite spins

**electronic configuration** (of an atom) the number of electrons in each sub-shell in each energy level of the atom

**Hund's rule** electrons will occupy the orbitals singly before pairing takes place

**Pauli Exclusion Principle** two electrons cannot occupy the same orbital unless they have opposite spins; electron spin is usually shown by using upward and downward arrows: ↑ and ↓

# 2A 4 IONISATION ENERGIES

SPECIFICATION REFERENCE: 2.8 2.10 2.11

## LEARNING OBJECTIVES

- Be able to define first, second, and third ionisation energies and understand that all ionisation energies are endothermic.
- Understand how ionisation energies are influenced by the number of protons in the nucleus, the electron shielding and the sub-shell from which the electron is removed.
- Know that ideas about electronic configurations developed from an understanding that: successive ionisation energies provide evidence for the existence of quantum shells and the group to which the element belongs; the first ionisation energy of successive elements provides evidence for electron sub-shells.
- Be able to represent ionisation energy data in graphical form.
- Explain the general increase in first ionisation energy across a period.
- Explain the decrease in first ionisation energy down a group.

When you studied chemistry previously, you probably learned that electrons exist in shells surrounding the nucleus. You may not have been told how we know this.

There is lots of experimental evidence to support this theory, including data from considering the ionisation energies of elements.

## IONISATION ENERGY

*Ionisation energy* is a measure of the energy required to completely remove an electron from an atom of an element.

We can represent the **first ionisation energy** of an element, A, by the equation:

$$A(g) \rightarrow A^+(g) + e^-$$

We can represent the **second ionisation energy** of A by the equation:

$$A^+(g) \rightarrow A^{2+}(g) + e^-$$

We can represent the **third ionisation energy** of A by the equation:

$$A^{2+}(g) \rightarrow A^{3+}(g) + e^-$$

## SUCCESSIVE IONISATION ENERGIES

When successive energies of an element are listed, there are steady increases, and big jumps occur at defined places. This is one piece of evidence for the existence of quantum shells.

A sodium (Na) atom has 11 electrons. Its successive ionisation energies are shown in **table A**. The places where the ionisation energy has jumped significantly are shown in red.

| 1st | 2nd | 3rd | 4th | 5th | 6th |
|---|---|---|---|---|---|
| 496 | 4563 | 6913 | 9544 | 13 352 | 16 611 |

| 7th | 8th | 9th | 10th | 11th |
|---|---|---|---|---|
| 20 115 | 25 491 | 28 934 | 141 367 | 159 079 |

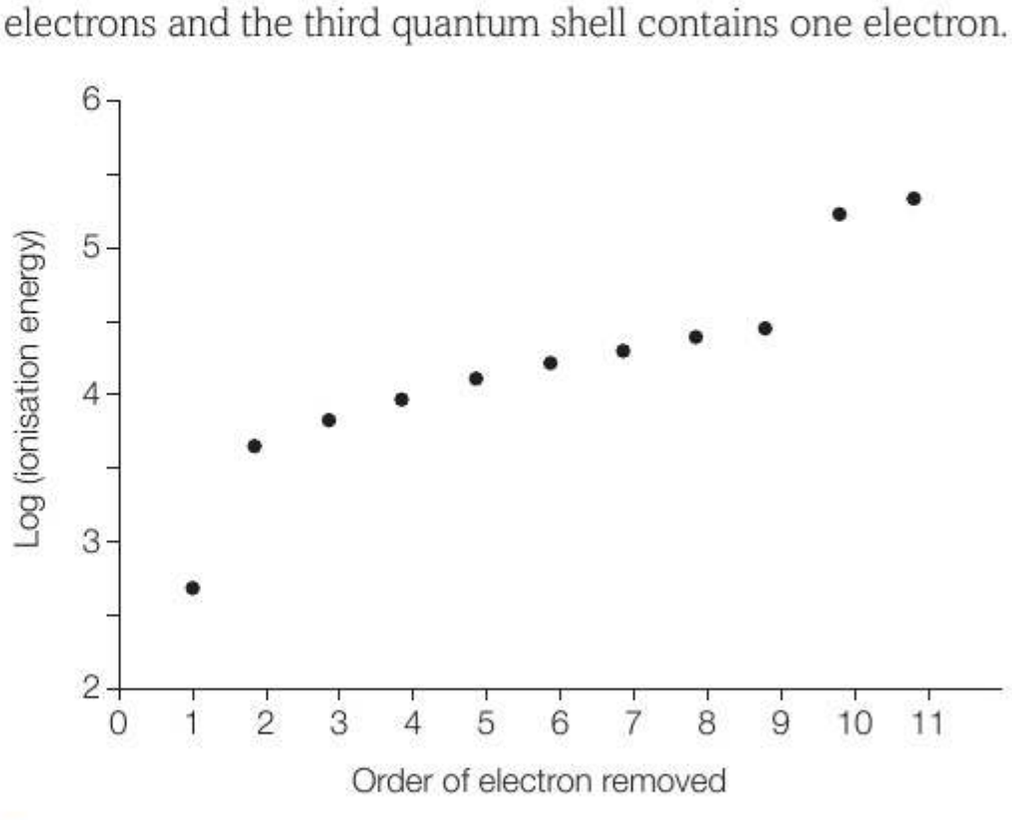
**table A** Successive ionisation energies, in kJ $mol^{-1}$, of sodium.

The first electron is considerably easier to remove than the second. There is a steady rise in ionisation energy for the next eight electrons. Then there is a big jump in energy from the ninth electron to the tenth. The last two electrons are much more difficult to remove than the previous eight.

The explanation for this trend is that the last two electrons to be removed are in the first quantum shell, the one that has the lowest energy. The next eight electrons are in the second quantum shell, which is of lower energy than the third. The first electron to be removed is in the third quantum shell of highest energy. This means that the basic electronic configuration of a sodium atom is 2,8,1.

The trend in successive ionisation energies can be seen when the logarithm of the ionisation energy is plotted against the order of electron removal. **Fig A** shows the graph for a sodium atom.

It is easy to see from the graph that there are three distinct quantum shells containing electrons. The first quantum shell contains two electrons, the second quantum shell contains eight electrons and the third quantum shell contains one electron.




▲ **fig A** Graph showing the trend in successive ionisation energies for sodium.

**EXAM HINT**

In an exam, it is worth quickly sketching a graph of successive ionisation energies, so that you can clearly see the pattern.

## WHY DO SUCCESSIVE IONISATION ENERGIES INCREASE IN MAGNITUDE?

We need to understand what ionisation energy represents so that we can answer this question.

The electron lost during ionisation is so far away from the influence of the nucleus that it no longer experiences an attractive force from the nucleus. We can say it is at an infinite distance from the nucleus. The energy of the electron has to be increased to a particular value for it to be removed. For any given atom, the energy value that the electron has when it reaches this position will always be the same, regardless of where in the atom the electron has come from.

If an electron already has a high energy, then it will not need to gain much energy to be removed. If, however, the electron is in an orbital of a low-energy quantum shell, for example the 1s orbital, then it will need to gain a lot more energy to be removed. The *difference* in energy between the electron when it has been removed and the energy it has when it is in its original orbital in the quantum shell is known as the ionisation energy. We can represent this by the equation:

Ionisation energy (IE) = energy of electron when removed − energy of electron when in the orbital

The ionisation energy for a particular electron in a given atom depends solely on the energy it has when it is in its orbital within the atom.

## WHY DO THE SUCCESSIVE IONISATION ENERGIES OF SODIUM INCREASE?

Look at **table A** showing the successive ionisation energies of sodium.

The first electron to be removed has the highest energy of any of the electrons in the sodium atom. It is in the third quantum shell in a 3s orbital. This means that the amount of energy it has to gain in order to be removed is the lowest for any of the electrons. The first ionisation energy is therefore the lowest of all the ionisation energies.

There is a large jump from the first to the second ionisation energy. This is because the second electron to be removed is in a quantum shell of considerably lower energy, the second quantum shell. There is a steady rise in ionisation energy from the second to the ninth electron. This indicates that the eight electrons all exist within the same quantum shell. As each successive electron is removed from this shell, the electron–electron repulsion within the shell decreases. This results in a decrease in the energies of the remaining electrons, and therefore a steady increase in ionisation energy from the second to the ninth electron.

The large jump from the ninth to the tenth electron indicates another significant change in energy of the electron. This corresponds to the removal of an electron from the first quantum shell: the 1s orbital. The tenth ionisation energy is larger than the ninth; the reason for this is that as one electron is removed from the 1s orbital, the remaining electron now experiences zero repulsion, so its energy decreases.

**LEARNING TIP**

Remember that electron–electron repulsion:

- exists between two electrons that are in the same orbital
- exists between electrons in different orbitals within a given quantum shell
- is sometimes most significant between electrons in adjacent quantum shells
- is sometimes called shielding or screening.

**DID YOU KNOW?**

**The terms 'shielding' and 'screening' are in some ways an unfortunate choice of words for the effect that electron–electron repulsion has on the energies of the electrons in their respective orbitals. The terms seem to suggest that, for example, the electrons in an inner quantum shell will set up a barrier to the attractive force of the nucleus to those electrons in an outer quantum shell. This is not necessarily the case. The effect of electron–electron repulsion is to raise the energy of the electrons involved above the value they would have if there was no repulsion between them. This then affects the amount of energy required to remove the electron from the atom or ion. That is, it affects the magnitude of the ionisation energy.**

## WHAT DETERMINES THE ENERGY OF AN ELECTRON?

You might think the answer to this question is very straightforward. You might think it is determined by the orbital in which the electron has been placed. As you will often find in chemistry, there are a number of different factors that affect the answer.

### HYDROGEN AND HELIUM

First, consider the atoms of hydrogen (H) and helium (He).

Their electronic configurations are below.

H $1s^1$

He $1s^2$

The outer electrons of both atoms are in a 1s orbital, so you might think that they will have the same energy. However, the 1s orbital of helium contains two electrons. This increases the electron–electron repulsion within the orbital: each electron shields the other from the effect of the nuclear charge. The effect of this factor alone would increase the energy of the electrons.

However, the nuclear charge of helium is double that of hydrogen. This is because it contains two protons as opposed to hydrogen's one proton. The effect of this increased nuclear charge is to decrease the energy of the 1s electrons since they are attracted more strongly.

In this case, the effect of the increased nuclear charge is greater than that of the increased shielding. This means that the first ionisation energy of helium (2370 kJ $mol^{-1}$) is larger than that of hydrogen (1310 kJ $mol^{-1}$).

### HELIUM AND LITHIUM

Now consider helium (He) and lithium (Li).

The electronic configuration of each is:

He $1s^2$

Li $1s^2 2s^1$

The nuclear charge of lithium is +3 (three protons), which is larger than the nuclear charge of helium (+2).

However, the outer electron of lithium is in the second quantum shell in a 2s orbital. The second quantum shell is at a higher energy level than the first quantum shell. Added to this, the electron in the 2s orbital experiences repulsion from the two inner 1s electrons, i.e. it experiences shielding from the nuclear charge.

The last two effects are more significant than the increased nuclear charge. This means that the first ionisation energy of lithium (519 kJ $mol^{-1}$) is smaller than the first ionisation energy of helium (2370 kJ $mol^{-1}$).

### SUMMARY

The factors that affect the energy of an electron are:

- the orbital in which the electron exists
- the nuclear charge of the atom (i.e. the number of protons in the nucleus)
- the repulsion (shielding) experienced by the electron from all the other electrons present.

## TRENDS IN IONISATION ENERGIES

The two major trends in ionisation energies in the Periodic Table are found:

- across a period
- down a group.

### ACROSS A PERIOD

To illustrate this trend, consider Period 2; and the elements lithium (Li) to neon (Ne).

As we move from Li to Ne across Period 2, the nuclear charge increases as the number of protons increases. This, on its own, would lead to an increased attraction between the nucleus and the electron, and therefore a decrease in the energy of the outermost electron, and an increase in first ionisation energy.

But, counteracting this, one more electron is added to the same quantum shell on each occasion and this increases the electron–electron repulsion within the quantum shell. This, on its own, would cause an increase in energy of the outermost electron, and would lead to a decrease in first ionisation energy.

The increase in nuclear charge is more significant than the increase in electron–electron repulsion. So there is a general increase in first ionisation energy across Period 2.

We can say the same of the Period 3 elements, sodium (Na) to argon (Ar), excluding the d-block elements, scandium (Sc) to zinc (Zn).

**LEARNING TIP**

This trend is not perfect and there are two exceptions in each of Periods 2 and 3: in Period 2, beryllium and boron (Be and B); and in Period 3, magnesium and aluminium (Mg and Al).

### DOWN A GROUP

To illustrate this trend, consider the Group 1 elements, lithium (Li) to caesium (Cs).

As we descend through Group 1 from Li to Cs, the nuclear charge increases as the number of protons increases. This would lead to an increased attraction between the nucleus and the electron, and therefore a decrease in energy of the outer electron. This, in turn, would lead to an increase in the first ionisation energy.

However, one new quantum shell is added on each occasion. This increases the energy of the outermost electron for two reasons. Firstly, the third quantum shell has a higher energy value than the second; the fourth quantum shell is higher in energy than the third.

This continues down the group. Secondly, as each new quantum shell is added, the outer electron experiences increased repulsion (i.e. increased shielding) from the inner electrons.

On this occasion, the combined effect of adding an extra shell and increasing the shielding is more significant than the increase in nuclear charge. So the first ionisation energy decreases down Group 1 from Li to Cs.

This trend is repeated in:

- Group 2: beryllium to barium (Be to Ba)
- Group 5: nitrogen to bismuth (N to Bi)
- Group 6: oxygen to polonium (O to Po)
- Group 7: fluorine to astatine (F to At)
- Group 8: neon to radon (Ne to Rn).

In Group 4, lead (Pb) is an anomaly (does not fit in with the others) because it has a first ionisation energy that is higher than that of tin (Sn), the element immediately above it.

There is no general trend in first ionisation energy in Group 3 (boron (B) to thallium (Tl)).

The explanations for the anomalies in Groups 3 and 4 are not included in this book, but they illustrate that, in chemistry, you can rarely apply a simple pattern or trend to all situations.

## CHECKPOINT

**1.** State the three factors that determine the magnitude of the first ionisation energy of an element.

**2.** Write equations to represent:

(a) the first ionisation energy of sodium

(b) the second ionisation energy of calcium

(c) the third ionisation energy of carbon.

**3.** Draw a sketch graph for the logarithm to base 10 for the successive ionisation energies of phosphorus.

**4.** The table below shows the first four ionisation energies, in kJ $mol^{-1}$, of five elements: A, B, C, D and E.

| ELEMENT | FIRST IONISATION ENERGY | SECOND IONISATION ENERGY | THIRD IONISATION ENERGY | FOURTH IONISATION ENERGY |
|---|---|---|---|---|
| **A** | 496 | 4563 | 6913 | 9544 |
| **B** | 738 | 1451 | 7733 | 10 541 |
| **C** | 578 | 1817 | 2745 | 11 578 |
| **D** | 900 | 1757 | 14 849 | 21 007 |
| **E** | 631 | 1235 | 2389 | 7089 |

(a) Which two elements are in the same group of the Periodic Table? Explain your answer.

(b) In which group of the Periodic Table is element C likely to occur? Explain your answer.

(c) Which element requires the least amount of energy to form a 2+ ion? Explain your answer.

**5.** The first four ionisation energies, in kJ $mol^{-1}$, of calcium are 590, 1145, 4912 and 6474.

(a) Explain why the second ionisation energy of calcium is larger than the first.

(b) Explain why the third ionisation energy is much larger than the second.

## SUBJECT VOCABULARY

**first ionisation energy** (of an element) the energy required to remove an electron from each atom in one mole of atoms in the gaseous state

**second ionisation energy** (of an element) the energy required to remove an electron from each singly charged positive ion in one mole of positive ions in the gaseous state

**third ionisation energy** (of an element) the energy required to remove an electron from each doubly charged positive ion in one mole of positive ions in the gaseous state